AI502 – MACHINE LEARNING

COMPREHENSIVE QUESTION BANK & SELF-ASSESSMENT COMPANION

Unit V (Unsupervised Learning & Basics of Neural Networks) & Unit VI (Advanced Learning Topics)

Total Questions: 106 MCQs | **Target Exam:** Semester Examination - 2 | **Format:** 4 Options + Verified Key & Detailed Explanation

UNIT V: UNSUPERVISED LEARNING & NEURAL NETWORKS

Q1. What is the fundamental distinction between Supervised Learning and Unsupervised Learning? [Unsupervised Learning Fundamentals | Easy]

A) Supervised learning trains on labeled input-output pairs (x, y); Unsupervised learning discovers latent structures in unlabeled data (x only)

B) Supervised learning requires GPU hardware; Unsupervised learning runs only on CPUs

C) Supervised learning is only used for image processing; Unsupervised learning is for text only

D) Unsupervised learning never uses mathematical loss functions

✓ Correct Answer: Supervised learning trains on labeled input-output pairs (x, y); Unsupervised learning discovers latent structures in unlabeled data (x only)

Explanation: In unsupervised learning, there are no predefined supervisor targets or ground-truth class labels provided during training.

Q2. Which of the following tasks is a classic application of Unsupervised Machine Learning? [Unsupervised Learning Fundamentals | Easy]

A) Customer segmentation and grouping based on purchasing behaviors

B) Predicting house sale prices from square footage (regression)

C) Classifying emails into spam or non-spam using labeled examples

D) Predicting whether a patient has diabetes based on medical ground truth

✓ Correct Answer: Customer segmentation and grouping based on purchasing behaviors

Explanation: Customer segmentation groups unlabeled customer records into clusters based on natural feature similarities.

Q3. What is the objective of partition-based clustering algorithms such as k-Means and k-Medoids? [Clustering & Partition Methods | Medium]

A) To partition n unlabeled data points into k distinct clusters such that intra-cluster similarity is maximized and inter-cluster similarity is minimized

B) To arrange data points into a continuous binary decision tree with infinite depth

C) To compute regression weights that fit a linear hyperplane

D) To generate artificial synthetic images from noise vectors

✓ Correct Answer: To partition n unlabeled data points into k distinct clusters such that intra-cluster similarity is maximized and inter-cluster similarity is minimized

Explanation: Partitioning methods divide datasets into k non-overlapping groups optimizing an objective function (e.g. inertia or sum of squared errors).

Q4. How does the k-Medoids (PAM - Partitioning Around Medoids) algorithm fundamentally differ from standard k-Means? [Clustering & Partition Methods | Hard]

A) k-Medoids chooses actual representative data points from the dataset as cluster centers, whereas k-Means computes the mathematical mean (centroid)

B) k-Medoids can only run on labeled categorical data

C) k-Means is immune to outliers while k-Medoids is heavily distorted by outliers

D) k-Medoids requires training a deep convolutional neural network

✓ Correct Answer: k-Medoids chooses actual representative data points from the dataset as cluster centers, whereas k-Means computes the mathematical mean (centroid)

Explanation: By selecting real exemplars (medoids) instead of virtual averages (means), k-Medoids is significantly more robust to noise and outliers.

Q5. What common algorithm is used to execute k-Medoids clustering? [Clustering & Partition Methods | Medium]

A) PAM (Partitioning Around Medoids)

B) Backpropagation algorithm

C) Apriori algorithm

D) Dijkstra algorithm

✓ Correct Answer: PAM (Partitioning Around Medoids)

Explanation: PAM is the classic standard algorithm for k-Medoids, evaluating whether swapping a medoid with a non-medoid point reduces total dissimilarity.

Q6. What is an inherent limitation of partition-based clustering algorithms like k-Means and k-Medoids? [Clustering & Partition Methods | Medium]

A) The number of clusters k must be specified in advance by the user, and they struggle with non-spherical clusters

B) They cannot run on numerical floating-point vectors

C) They can only process datasets containing fewer than 50 rows

D) They require labeled test sets to converge

✓ Correct Answer: The number of clusters k must be specified in advance by the user, and they struggle with non-spherical clusters

Explanation: Choosing an optimal k requires heuristic methods (e.g. Elbow method or Silhouette score), and partitions assume convex spherical geometry.

Q7. What graphical tree diagram is produced by Hierarchical Clustering to visualize nested cluster relationships? [Hierarchical & Density Clustering | Easy]

A) Dendrogram

B) Histogram

C) Scatter plot matrix

D) Confusion Matrix

✓ Correct Answer: Dendrogram

Explanation: A dendrogram illustrates the hierarchical sequence of nested merges (agglomerative) or splits (divisive) across various distance thresholds.

Q8. What is the difference between Agglomerative and Divisive hierarchical clustering? [Hierarchical & Density Clustering | Medium]

A) Agglomerative is bottom-up (starts with individual points and merges clusters); Divisive is top-down (starts with one cluster and splits)

B) Agglomerative is supervised; Divisive is unsupervised

C) Agglomerative operates only on images; Divisive operates only on audio

D) Agglomerative requires defining k in advance, whereas Divisive does not

✓ Correct Answer: Agglomerative is bottom-up (starts with individual points and merges clusters); Divisive is top-down (starts with one cluster and splits)

Explanation: Agglomerative bottom-up clustering starts with N clusters and iteratively joins the closest pair until one cluster remains.

Q9. What is the core principle of Density-Based Clustering algorithms such as DBSCAN? [Hierarchical & Density Clustering | Hard]

A) Clusters are defined as contiguous dense regions of points in data space separated by regions of low density (noise)

B) Clusters are defined by drawing linear hyperplanes through the global origin

C) Data points are sorted alphabetically and partitioned into equal quartiles

D) Clusters must always form perfect geometric hyper-spheres

✓ Correct Answer: Clusters are defined as contiguous dense regions of points in data space separated by regions of low density (noise)

Explanation: DBSCAN discovers clusters of arbitrary shapes and isolates outliers as noise based on density criteria (Eps and MinPts).

Q10. What are the two key user parameters required by the DBSCAN density clustering algorithm? [Hierarchical & Density Clustering | Medium]

A) Epsilon (Eps, neighborhood radius) and MinPts (minimum points in neighborhood)

B) Learning rate and Momentum

C) Number of clusters k and maximum tree depth

D) Kernel size and pooling stride

✓ Correct Answer: Epsilon (Eps, neighborhood radius) and MinPts (minimum points in neighborhood)

Explanation: A point is a core point in DBSCAN if at least MinPts points reside within its radial distance ϵ (Eps).

Q11. What is the primary goal of Association Rule Mining in unsupervised learning (e.g. Market Basket Analysis)? [Association Rule Learning | Medium]

A) To discover interesting correlations, frequent patterns, and co-occurrence relationships among items in large transaction datasets

B) To minimize the mean squared error of a continuous price curve

C) To assign class labels using a maximum-margin hyperplane

D) To compress high-resolution video streams in real time

✓ Correct Answer: To discover interesting correlations, frequent patterns, and co-occurrence relationships among items in large transaction datasets

Explanation: Association rules (e.g. $\text{Bread} \Rightarrow \text{Butter}$) uncover probabilistic relationships between items purchased or used together.

Q12. In association rule mining, what does the metric 'Support' of an itemset $\{A, B\}$ measure? [Association Rule Learning | Hard]

A) The proportion of total transactions in the dataset that contain both item A and item B

B) The conditional probability that item B is purchased given that item A was purchased

C) The ratio of the observed co-occurrence to the expected independence rate

D) The price difference between item A and item B

✓ Correct Answer: The proportion of total transactions in the dataset that contain both item A and item B

Explanation: Support measures the statistical frequency or popularity of an itemset across all transactions: $\text{Support}(X) = \text{count}(X) / N$.

Q13. In an association rule $A \Rightarrow B$, what does the metric 'Confidence' measure? [Association Rule Learning | Hard]

A) The conditional probability $P(B|A)$ that item B is present given that item A is present in a transaction

B) The total percentage of all transactions that contain item A

C) The correlation coefficient between the weights of item A and item B

D) The probability that neither item A nor item B appears in the database

✓ Correct Answer: The conditional probability $P(B|A)$ that item B is present given that item A is present in a transaction

Explanation: Confidence gauges rule reliability: $\text{Confidence}(A \Rightarrow B) = \text{Support}(A \cup B) / \text{Support}(A)$.

Q14. What classic algorithm utilizes the downward closure property (apriori property) that all subsets of a frequent itemset must also be frequent? [Association Rule Learning | Medium]

A) Apriori Algorithm

B) Dijkstra Algorithm

C) k-Medoids Algorithm

D) Backpropagation Algorithm

✓ **Correct Answer: Apriori Algorithm**

Explanation: The Apriori algorithm prunes candidate itemsets by utilizing the property: if an itemset is infrequent, all its supersets must be infrequent.

Q15. In the biological nervous system, what biological structures correspond roughly to the inputs, weights, cell body, and output of an artificial neuron? [Neural Network Fundamentals | Medium]

A) Dendrites (inputs), Synaptic strengths (weights), Soma (cell body/summation), and Axon (output)

B) Axons (inputs), Dendrites (weights), Nucleus (summation), and Myelin (output)

C) Cortex (inputs), Neurons (weights), Cerebellum (summation), and Skull (output)

D) Ventricles (inputs), Spine (weights), Nerves (summation), and Muscles (output)

✓ **Correct Answer: Dendrites (inputs), Synaptic strengths (weights), Soma (cell body/summation), and Axon (output)**

Explanation: Dendrites receive incoming signals, synapses weight them, the soma integrates them, and the axon transmits the resulting activation.

Q16. What early mathematical model of an artificial neuron was proposed in 1943 by Warren McCulloch and Walter Pitts? [Neural Network Fundamentals | Easy]

A) McCulloch-Pitts (MCP) Neuron

B) Transformer Attention Head

C) Hopfield Network

D) Boltzmann Machine

✓ **Correct Answer: McCulloch-Pitts (MCP) Neuron**

Explanation: The McCulloch-Pitts neuron was the pioneering threshold logic unit capable of computing basic boolean functions (AND, OR, NOT).

Q17. What key improvement did Frank Rosenblatt's Perceptron (1958) introduce over the original McCulloch-Pitts model? [Neural Network Fundamentals | Medium]

A) Learnable, dynamically adjustable numerical synaptic weights and a systematic learning algorithm

B) Multi-head self-attention mechanisms

C) Automatic generation of labeled training sets

D) Support for non-linear XOR separation without hidden layers

✓ **Correct Answer: Learnable, dynamically adjustable numerical synaptic weights and a systematic learning algorithm**

Explanation: Rosenblatt's perceptron introduced an iterative weight update rule that guaranteed convergence for linearly separable patterns.

Q18. What does the ADALINE (Adaptive Linear Neuron) model developed by Widrow and Hoff use for weight updates? [Neural Network Fundamentals | Hard]

A) The Delta Rule (Least Mean Squares / LMS / Widrow-Hoff learning rule) based on continuous linear outputs before thresholding

B) Backpropagation through 50 hidden layers

C) Genetic crossover algorithms

D) K-nearest neighbor voting

✓ **Correct Answer: The Delta Rule (Least Mean Squares / LMS / Widrow-Hoff learning rule) based on continuous linear outputs before thresholding**

Explanation: ADALINE updates weights using the difference between desired target and analog pre-activation linear sum, minimizing Mean Squared Error.

Q19. Why are non-linear activation functions necessary in multi-layer neural networks? [Activation Functions | Hard]

A) Without non-linearities, a multi-layer network collapses mathematically into an equivalent single-layer linear model unable to learn complex patterns

B) Because linear functions consume too much computer RAM

C) Because CPUs cannot perform matrix addition

D) To prevent numbers from becoming negative integers

✓ **Correct Answer: Without non-linearities, a multi-layer network collapses mathematically into an equivalent single-layer linear model unable to learn complex patterns**

Explanation: A composition of linear transformations is strictly another linear transformation; non-linearities enable universal function approximation.

Q20. What is the mathematical formulation of the Sigmoid activation function, and what is its output range? [Activation Functions | Medium]

A) $\sigma(z) = \frac{1}{1 + e^{-z}}$, producing outputs in the open interval $(0, 1)$

B) $\sigma(z) = \max(0, z)$, producing outputs in $[0, \infty)$

C) $\sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, producing outputs in $(-1, 1)$

D) $\sigma(z) = z$, producing outputs in $(-\infty, \infty)$

✓ Correct Answer: $\sigma(z) = \frac{1}{1 + e^{-z}}$, producing outputs in the open interval $(0, 1)$

Explanation: Sigmoid squashes real values into probabilities between 0 and 1, commonly used for binary classification output nodes.

Q21. What is the formula and output range of the Hyperbolic Tangent (tanh) activation function? [Activation Functions | Medium]

A) $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, producing zero-centered outputs in the interval $(-1, 1)$

B) $\tanh(z) = \frac{1}{1 + e^{-z}}$, producing outputs in $(0, 1)$

C) $\tanh(z) = \max(0, z)$, producing outputs in $[0, 1]$

D) $\tanh(z) = e^z$, producing outputs in $[1, \infty)$

✓ Correct Answer: $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, producing zero-centered outputs in the interval $(-1, 1)$

Explanation: tanh outputs are zero-centered, which often aids gradient descent convergence compared to the strictly positive sigmoid function.

Q22. What is the formula and primary advantage of the Rectified Linear Unit (ReLU) activation function? [Activation Functions | Medium]

A) $f(z) = \max(0, z)$; it is computationally simple and alleviates the vanishing gradient problem in deep networks

B) $f(z) = 1 / (1 + e^{-z})$; it squashes outputs to zero

C) $f(z) = z^2$; it makes all negative values positive

D) $f(z) = \sin(z)$; it creates periodic smooth outputs

✓ Correct Answer: $f(z) = \max(0, z)$; it is computationally simple and alleviates the vanishing gradient problem in deep networks

Explanation: ReLU has a constant derivative of 1 for $z > 0$, preventing gradients from vanishing exponentially during backpropagation in deep architectures.

Q23. Which activation function is standardly used in the final layer of a multi-class classification neural network to output normalized class probabilities? [Activation Functions | Medium]

A) Softmax function (sums to 1 across all classes)

B) Identity / Linear function

C) Binary Step function

D) Hard Sigmoid function

✓ Correct Answer: Softmax function (sums to 1 across all classes)

Explanation: Softmax normalizes real-valued logits into a categorical probability distribution where all class probabilities sum to exactly 1.0.

Q24. What is a Single-Layer Feed-Forward Network, and what historic limitation did Minsky and Papert prove in 1969? [Network Architectures | Hard]

A) A network with one input layer and one output layer; it cannot solve non-linearly separable problems such as the XOR logic function

B) A network that loops outputs back into inputs indefinitely

C) A network that cannot compute the AND or OR logic functions

D) A network that requires quantum computers to train

✓ Correct Answer: A network with one input layer and one output layer; it cannot solve non-linearly separable problems such as the XOR logic function

Explanation: Single-layer perceptrons can only classify linearly separable classes; solving XOR required hidden layers (Multi-Layer Perceptrons).

Q25. What algorithm enabled the efficient training of Multi-Layer Feed-Forward Networks by computing error gradients layer by layer? [Network Architectures | Hard]

A) Backpropagation (Backpropagation of Error using the mathematical Chain Rule of calculus)

B) Apriori candidate generation

C) DBSCAN density expansion

D) Bubble sort gradient descent

✓ Correct Answer: Backpropagation (Backpropagation of Error using the mathematical Chain Rule of calculus)

Explanation: Backpropagation propagates output errors backward through hidden layers via the calculus chain rule, calculating partial derivatives with respect to each weight.

Q26. What is a Recurrent Neural Network (RNN)? [Network Architectures | Medium]

A) A neural network containing directed cycles in connections, providing internal memory to process sequential/time-series data

B) A network that discards all previous inputs and operates only on static images

C) A network with strictly feed-forward connections that cannot process text

D) A hardware processor that runs without electricity

✓ Correct Answer: A neural network containing directed cycles in connections, providing internal memory to process sequential/time-series data

Explanation: RNNs maintain a hidden recurrent state that retains information across time steps, making them suited for sequence modeling.

Q27. What key architectural components enable Convolutional Neural Networks (CNNs) to excel in computer vision and image classification? [Network Architectures | Hard]

A) Convolutional filters with local receptive fields, parameter (weight) sharing, and pooling layers for spatial invariance

B) Fully recurrent temporal feedback loops at every pixel

C) Association rule pruning using minimum confidence thresholds

D) Hierarchical agglomerative dendrogram generation

✓ Correct Answer: Convolutional filters with local receptive fields, parameter (weight) sharing, and pooling layers for spatial invariance

Explanation: CNNs apply learnable spatial filters across feature maps, extracting hierarchical visual features (edges, textures, shapes) efficiently.

Q28. What revolutionary mechanism powers modern Transformer architectures (such as GPT and BERT) to process long-range dependencies in parallel? [Network Architectures | Hard]

A) Self-Attention Mechanism (Scaled Dot-Product Self-Attention)

B) Strictly sequential recurrent state propagation

C) k-Medoids medoid swapping

D) Markov Decision Processes without weights

✓ Correct Answer: Self-Attention Mechanism (Scaled Dot-Product Self-Attention)

Explanation: Transformers dispense with recurrence, computing attention scores between all token positions in parallel to model global dependencies.

Q29. What is a Competitive Neural Network (such as Kohonen's Self-Organizing Map / SOM)? [Network Architectures | Hard]

A) An unsupervised network where neurons compete to be activated (Winner-Take-All), clustering similar input vectors onto a topological grid

B) A network where two models predict stock market prices against each other

C) A network that evaluates chess moves using minimax trees

D) A multi-layer perceptron trained with cross-entropy loss

✓ Correct Answer: An unsupervised network where neurons compete to be activated (Winner-Take-All), clustering similar input vectors onto a topological grid

Explanation: In competitive learning, the neuron whose weight vector is closest to the input is declared the winner and updates its weights.

UNIT VI: ADVANCED LEARNING TOPICS

Q30. What is Representation Learning (or Feature Learning) in modern machine learning? [Advanced Learning Paradigms | Medium]

A) Learning representations of the data that make it easier to extract useful information when building classifiers or other predictors automatically

B) Manually hand-crafting thousands of CSV columns in Excel

C) Translating source code into assembly language

D) Printing machine learning models on paper

✓ Correct Answer: Learning representations of the data that make it easier to extract useful information when building classifiers or other predictors automatically

Explanation: Representation learning automatically discovers optimal feature transformations directly from raw data (e.g. deep autoencoders and embeddings).

Q31. What is the core premise of Active Learning in machine learning? [Advanced Learning Paradigms | Hard]

A) The learning algorithm interactively queries a human oracle to label only the most informative or uncertain unlabeled data points

B) Training machine learning models while running physical treadmill exercises

C) Allowing neural networks to modify their own Python source code

D) Discarding all training data and using random guesses

✓ Correct Answer: The learning algorithm interactively queries a human oracle to label only the most informative or uncertain unlabeled data points

Explanation: Active learning dramatically reduces labeling costs by strategically asking experts to label instances where the model has highest uncertainty.

Q32. What is Instance-Based Learning (also known as Lazy Learning)? [Advanced Learning Paradigms | Medium]

A) Algorithms that delay generalization until a query is made, comparing new instances directly with stored training examples (e.g. k-NN)

B) Algorithms that train deep neural networks instantly in zero seconds

C) Algorithms that require no computer memory to run

D) Algorithms that can only run once before self-destructing

✓ Correct Answer: Algorithms that delay generalization until a query is made, comparing new instances directly with stored training examples (e.g. k-NN)

Explanation: In contrast to eager learners (which build an explicit model during training), lazy learners store instances and evaluate distances at query time.

Q33. In the k-Nearest Neighbors (k-NN) instance-based algorithm, what is the effect of choosing a very small value of k (e.g. $k = 1$)? [Advanced Learning Paradigms | Hard]

A) High model complexity with low bias and high variance (prone to overfitting and sensitive to noise)

B) High bias with an excessively smooth decision boundary (underfitting)

C) Zero execution time during inference

D) The model becomes identical to a linear regression line

✓ Correct Answer: High model complexity with low bias and high variance (prone to overfitting and sensitive to noise)

Explanation: $k=1$ captures local noise and creates fragmented boundaries (high variance); larger k produces smoother, more generalized boundaries.

Q34. What is the core concept of Ensemble Learning algorithms in machine learning? [Ensemble Learning | Easy]

A) Combining the predictions of multiple base models (weak learners) to produce a single superior model with lower variance or bias

B) Training a single perceptron using only 1 data row

C) Converting a neural network into an association rule table

D) Using multiple monitors while coding Python scripts

✓ Correct Answer: Combining the predictions of multiple base models (weak learners) to produce a single superior model with lower variance or bias

Explanation: Ensemble methods (Bagging, Boosting, Stacking) harness the collective wisdom of an ensemble of models to outperform any individual constituent model.

Q35. What is Bagging (Bootstrap Aggregating) in ensemble learning, and what is its primary benefit? [Ensemble Learning | Hard]

A) Training multiple base models in parallel on independent bootstrap samples of the data and averaging their predictions to reduce variance (e.g. Random Forest)

B) Training models sequentially to correct prior mistakes and reduce bias

C) Compressing datasets using lossy JPEG algorithms

D) Pruning decision trees until only one leaf node remains

✓ Correct Answer: Training multiple base models in parallel on independent bootstrap samples of the data and averaging their predictions to reduce variance (e.g. Random Forest)

Explanation: Bagging trains models on random subsets with replacement; averaging predictions stabilizes erratic estimators and reduces overfitting.

Q36. What is Boosting (e.g. AdaBoost, Gradient Boosting, XGBoost) in ensemble learning? [Ensemble Learning | Hard]

A) An ensemble technique where base models are trained sequentially, with each new model focusing on correcting the errors made by previous models

B) Running machine learning scripts with elevated CPU overclocking

C) Averaging predictions from 100 identical linear regressions

D) Normalizing data features between 0 and 1

✓ Correct Answer: An ensemble technique where base models are trained sequentially, with each new model focusing on correcting the errors made by previous models

Explanation: Boosting builds models sequentially, adaptively reweighting misclassified samples or fitting pseudo-residuals to reduce bias.

Q37. What is the primary purpose of Regularization algorithms in machine learning? [Regularization Algorithms | Medium]

A) To prevent overfitting by penalizing overly complex models and large weight parameters

B) To speed up database file downloading over Wi-Fi

C) To convert supervised models into unsupervised clustering models

D) To eliminate the need for an activation function in neural networks

✓ Correct Answer: To prevent overfitting by penalizing overly complex models and large weight parameters

Explanation: Regularization adds a penalty term to the loss function ($J(\theta) + \lambda \cdot \Omega(\theta)$) to discourage model complexity.

Q38. What penalty is added to the loss function in L1 Regularization (Lasso), and what distinctive feature does it produce? [Regularization Algorithms | Hard]

A) L1 adds the sum of absolute values of weights ($\lambda \sum |w_i|$); it drives irrelevant feature weights strictly to zero, performing feature selection

B) L1 adds the sum of squared weights; it shrinks weights smoothly without zeroing

C) L1 multiplies weights by the learning rate

D) L1 removes all bias terms from the model

✓ Correct Answer: L1 adds the sum of absolute values of weights ($\lambda \sum |w_i|$); it drives irrelevant feature weights strictly to zero, performing feature selection

Explanation: Lasso (L1) yields sparse weight vectors, effectively eliminating uninformative features by setting their coefficients exactly to 0.

Q39. What penalty is added to the loss function in L2 Regularization (Ridge / Weight Decay)? [Regularization Algorithms | Hard]

A) L2 adds the sum of squared weights ($\lambda \sum w_i^2$); it shrinks weights close to zero, preventing any single feature from dominating

B) L2 adds the sum of absolute values of weights

C) L2 divides weights by the number of training samples

D) L2 sets all weights to positive integers

✓ Correct Answer: L2 adds the sum of squared weights ($\lambda \sum w_i^2$); it shrinks weights close to zero, preventing any single feature from dominating

Explanation: Ridge (L2) penalizes large weights proportionally to their square, distributing weight values smoothly and mitigating multicollinearity.

Q40. What popular neural network regularization technique randomly deactivates a subset of neurons during each forward/backward training pass? [Regularization Algorithms | Medium]

A) Dropout

B) Early Stopping

C) Data Augmentation

D) Batch Normalization

✓ Correct Answer: Dropout

Explanation: Dropout forces the network to learn redundant, robust representations by preventing neurons from co-adapting excessively to specific features.

Q41. What is the fundamental difference between Discriminative Models and Generative Models? [Generative Models | Hard]

A) Discriminative models model the conditional probability $P(y|x)$ to predict labels; Generative models model the joint distribution $P(x, y)$ or $P(x)$ to generate new data

B) Discriminative models generate realistic synthetic images; Generative models output binary numbers

C) Discriminative models are unsupervised; Generative models are supervised

D) Generative models can only calculate linear regression slopes

✓ Correct Answer: Discriminative models model the conditional probability $P(y|x)$ to predict labels; Generative models model the joint distribution $P(x, y)$ or $P(x)$ to generate new data

Explanation: Discriminative models find decision boundaries between classes, whereas generative models capture underlying data distributions to synthesize realistic new samples.

Q42. What two neural network models compete against each other in a Generative Adversarial Network (GAN) framework? [Generative Models | Hard]

A) A Generator (which synthesizes realistic fake data) and a Discriminator (which distinguishes real samples from generated fakes)

B) An Encoder and a Decoder

C) A Perceptron and an ADALINE

D) A Regressor and a Classifier

✓ Correct Answer: A Generator (which synthesizes realistic fake data) and a Discriminator (which distinguishes real samples from generated fakes)

Explanation: In a GAN, the generator strives to deceive the discriminator, while the discriminator strives to detect counterfeit samples in a minimax game.

Q43. What is a Variational Autoencoder (VAE)? [Generative Models | Hard]

A) A directed probabilistic generative model that encodes inputs into a continuous latent distribution and decodes samples back to data space

B) A lossy compression tool for MP3 audio files

C) A simple decision tree with variance pruning

D) A linear regression model that predicts continuous real values

✓ Correct Answer: A directed probabilistic generative model that encodes inputs into a continuous latent distribution and decodes samples back to data space

Explanation: VAEs regularize the latent space by penalizing divergence from a prior distribution (typically Gaussian using KL-divergence), enabling smooth generative sampling.

Q44. What is the foundational framework of Reinforcement Learning (RL)? [Reinforcement Learning | Medium]

A) An Agent interacts with an Environment through Actions, receives Observations and scalar Rewards, and learns a Policy to maximize cumulative reward

B) Training a model using a pre-labeled spreadsheet of inputs and outputs

C) Partitioning unlabeled customer data into k spherical clusters

D) Compressing text files using Huffman coding

✓ Correct Answer: An Agent interacts with an Environment through Actions, receives Observations and scalar Rewards, and learns a Policy to maximize cumulative reward

Explanation: RL does not rely on explicit input-output supervisory pairs; instead, an agent learns optimal behavior through trial-and-error reward signals.

Q45. What mathematical framework is standardly used to formalize discrete Reinforcement Learning problems?
[Reinforcement Learning | Hard]

A) Markov Decision Process (MDP defined by States, Actions, Transition Probabilities, Rewards, and Discount Factor γ)

B) Normal Gaussian Bell Curve

C) Boolean Algebra Truth Table

D) Linear Programming Simplex Method

✓ Correct Answer: Markov Decision Process (MDP defined by States, Actions, Transition Probabilities, Rewards, and Discount Factor γ)
Explanation: MDPs provide the mathematical foundation for RL, formalizing environments where outcomes are partly random and partly under agent control.

Q46. What is the fundamental dilemma between 'Exploration' and 'Exploitation' in Reinforcement Learning?
[Reinforcement Learning | Medium]

A) Balancing between trying unfamiliar actions to discover better future rewards (exploration) and executing known actions yielding high current reward (exploitation)

B) Balancing computer memory usage and CPU clock frequency

C) Balancing dataset downloading speed and storage capacity

D) Balancing keyboard typing speed and mouse scrolling

✓ Correct Answer: Balancing between trying unfamiliar actions to discover better future rewards (exploration) and executing known actions yielding high current reward (exploitation)
Explanation: If an agent only exploits known paths, it may miss superior strategies; if it only explores, it fails to capitalize on learned optimal behaviors.

Q47. What is Q-Learning in Reinforcement Learning? **[Reinforcement Learning | Hard]**

A) A model-free, off-policy temporal difference algorithm that learns the quality (Q-value) of taking an action in a given state

B) A supervised algorithm for image segmentation

C) A clustering technique that builds dendrogram trees

D) An algorithm that calculates shortest paths on network routers

✓ Correct Answer: A model-free, off-policy temporal difference algorithm that learns the quality (Q-value) of taking an action in a given state
Explanation: Q-Learning updates state-action values using the Bellman optimality equation: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.

UNIT V: UNSUPERVISED LEARNING & NEURAL NETWORKS

Q48. Which metric is commonly used in k-Means clustering to evaluate cluster compactness and determine the optimal number of clusters via the Elbow method? **[Clustering & Partition Methods | Medium]**

A) Within-Cluster Sum of Squares (WCSS) / Inertia

B) Binary Cross-Entropy Loss

C) Mean Absolute Percentage Error (MAPE)

D) Area Under the ROC Curve (AUC)

✓ Correct Answer: Within-Cluster Sum of Squares (WCSS) / Inertia
Explanation: The Elbow method plots WCSS against k; the point where the rate of decrease sharply shifts (elbow point) represents an optimal cluster tradeoff.

Q49. What is the Silhouette Coefficient and what does a value close to +1 indicate? **[Clustering & Partition Methods | Hard]**

A) A measure of how similar an object is to its own cluster compared to other clusters; +1 indicates highly dense and well-separated clusters

B) A measure of classification accuracy on training data; +1 indicates zero false positives

C) A measure of model parameter size; +1 indicates 1 million parameters

D) A test of gradient vanishing in recurrent neural networks

✓ Correct Answer: A measure of how similar an object is to its own cluster compared to other clusters; +1 indicates highly dense and well-separated clusters
Explanation: Silhouette score ranges from -1 to +1, where values near +1 indicate sample is far from neighboring clusters and tightly bound to its own.

Q50. What is the fundamental difference between Agglomerative and Divisive hierarchical clustering? [Hierarchical Clustering | Medium]

A) Agglomerative is a bottom-up approach merging pairs of clusters; Divisive is a top-down approach splitting clusters

B) Agglomerative uses neural networks; Divisive uses decision trees

C) Agglomerative requires predefined cluster counts k; Divisive does not

D) Agglomerative applies only to continuous 1D data; Divisive applies to multi-dimensional data

✓ Correct Answer: Agglomerative is a bottom-up approach merging pairs of clusters; Divisive is a top-down approach splitting clusters

Explanation: Agglomerative starts with each point as an individual singleton cluster and merges nearest clusters until one cluster remains.

Q51. What tree-like diagram is generated by hierarchical clustering to visualize nested clusters and merge distances? [Hierarchical Clustering | Easy]

A) Dendrogram

B) Histogram

C) Box-and-whisker plot

D) Scatter plot matrix

✓ Correct Answer: Dendrogram

Explanation: A dendrogram illustrates the hierarchical sequence of cluster joins or divisions, and vertical cuts indicate chosen cluster partitions.

Q52. In hierarchical clustering, what is 'Complete Linkage'? [Hierarchical Clustering | Hard]

A) Computing distance between two clusters as the maximum distance between any single point in Cluster A and any single point in Cluster B

B) Computing distance as the minimum distance between closest points (single linkage)

C) Computing distance as the distance between cluster centroids

D) Computing distance using random sampling of points

✓ Correct Answer: Computing distance between two clusters as the maximum distance between any single point in Cluster A and any single point in Cluster B

Explanation: Complete linkage measures the maximum pairwise distance, producing more compact, sphere-like clusters than single linkage.

Q53. In the DBSCAN algorithm, what is a 'Core Point'? [Density-Based Clustering | Medium]

A) A point that has at least MinPts within its Eps (epsilon) neighborhood radius

B) A point that has zero neighbors within its radius

C) The mathematical arithmetic mean of all data points

D) The point with the highest prediction confidence in a classifier

✓ Correct Answer: A point that has at least MinPts within its Eps (epsilon) neighborhood radius

Explanation: DBSCAN classifies points as Core points (dense neighborhood >= MinPts), Border points (reachable from core), or Noise (outliers).

Q54. What major advantage does DBSCAN possess over standard k-Means clustering? [Density-Based Clustering | Medium]

A) DBSCAN can discover arbitrarily shaped clusters and is naturally resilient to noise and outliers without requiring k in advance

B) DBSCAN is faster than O(1) constant time

C) DBSCAN is guaranteed to find global optimal hyperplane boundaries

D) DBSCAN does not require distance metric computation

✓ Correct Answer: DBSCAN can discover arbitrarily shaped clusters and is naturally resilient to noise and outliers without requiring k in advance

Explanation: k-Means assumes spherical clusters of equal size, whereas DBSCAN forms clusters around dense contiguous regions of arbitrary shape.

Q55. In association rule mining, what does the 'Lift' of an association rule A -> B represent? [Pattern Finding & Association Rules | Hard]

A) The ratio of observed joint confidence to expected confidence if A and B were completely statistically independent

B) The total number of transactions containing item A divided by total transactions

C) The percentage of transactions containing B that also contain A

D) The execution speed of the Apriori algorithm

✓ Correct Answer: The ratio of observed joint confidence to expected confidence if A and B were completely statistically independent

*Explanation: Lift = P(A and B) / (P(A) * P(B)). A lift > 1 indicates positive dependence; lift = 1 indicates independence; lift < 1 indicates negative correlation.*

Q56. What is the core pruning principle of the Apriori algorithm? [Pattern Finding & Association Rules | Hard]

A) If an itemset is infrequent, then all of its supersets are guaranteed to be infrequent and can be pruned immediately

B) All subsets of an infrequent itemset must be frequent

C) Frequent itemsets cannot contain more than two items

D) Only items with zero transaction frequency are evaluated

✓ Correct Answer: If an itemset is infrequent, then all of its supersets are guaranteed to be infrequent and can be pruned immediately

Explanation: The downward-closure (anti-monotonicity) property states that any subset of a frequent itemset must be frequent, allowing massive search space pruning.

Q57. What mathematical structure models the biological neuron's dendrites, cell body (soma), and axon in an Artificial Neuron? [Basics of Neural Networks | Easy]

A) Inputs with synaptic weights (dendrites), summation and bias (soma), and activation function output (axon)

B) Database tables (dendrites), SQL queries (soma), and CSV exports (axon)

C) RAM addresses (dendrites), CPU clock (soma), and Cache (axon)

D) Network packets (dendrites), router buffer (soma), and switch port (axon)

✓ Correct Answer: Inputs with synaptic weights (dendrites), summation and bias (soma), and activation function output (axon)

Explanation: The artificial neuron multiplies incoming signals by weights, sums them with a bias offset, and passes the sum through an activation function.

Q58. What was a key historical limitation of the single-layer McCulloch-Pitts neuron and Rosenblatt Perceptron? [Basics of Neural Networks | Medium]

A) They could only classify linearly separable functions and could not learn the non-linear XOR (Exclusive OR) function

B) They could not perform additions or subtractions

C) They required quantum computers to compute outputs

D) They could only process text strings, not numeric vectors

✓ Correct Answer: They could only classify linearly separable functions and could not learn the non-linear XOR (Exclusive OR) function

Explanation: Minsky and Papert demonstrated in 1969 that single-layer perceptrons cannot separate non-linear logical functions like XOR.

Q59. What is the mathematical definition and range of the Sigmoid activation function? [Basics of Neural Networks | Medium]

A) sigma(z) = 1 / (1 + exp(-z)), with output bounded between (0, 1)

B) sigma(z) = max(0, z), with output unbounded between [0, infinity)

C) sigma(z) = (exp(z) - exp(-z)) / (exp(z) + exp(-z)), with output between (-1, 1)

D) sigma(z) = z, with output in (-infinity, +infinity)

✓ Correct Answer: sigma(z) = 1 / (1 + exp(-z)), with output bounded between (0, 1)

Explanation: Sigmoid squashes input real values into the range (0, 1), making it ideal for modeling Bernoulli probabilities.

Q60. Why is the ReLU (Rectified Linear Unit) activation function preferred over Sigmoid in deep hidden layers? [Basics of Neural Networks | Medium]

A) ReLU avoids vanishing gradient problem for positive values (derivative is constant 1) and is computationally fast

B) ReLU restricts all activations strictly between -1 and +1

C) ReLU completely eliminates the need for backpropagation

D) ReLU converts multi-class predictions into text tokens automatically

✓ Correct Answer: ReLU avoids vanishing gradient problem for positive values (derivative is constant 1) and is computationally fast

Explanation: For positive inputs, ReLU has gradient 1, which prevents gradients from saturating and diminishing across deep layered architectures.

Q61. What activation function is standardly placed at the output layer of a multi-class neural network classifier to generate a normalized probability distribution across K classes? [Basics of Neural Networks | Easy]

A) Softmax

B) Step function

C) Identity function

D) Hyperbolic tangent (tanh)

✓ Correct Answer: Softmax

Explanation: Softmax exponentiates logits and divides each by the sum of exponentials, ensuring all output values are positive and sum exactly to 1.

Q62. What is ADALINE (Adaptive Linear Neuron) and what learning rule did Bernard Widrow introduce with it? [Basics of Neural Networks | Hard]

A) ADALINE uses continuous linear activations for weight updates via the Delta rule (Least Mean Squares / Widrow-Hoff learning rule)

B) ADALINE uses discrete hard-limiting step functions for gradient descent

C) ADALINE is an unsupervised clustering algorithm

D) ADALINE uses convolutional kernels to process audio signals

✓ Correct Answer: ADALINE uses continuous linear activations for weight updates via the Delta rule (Least Mean Squares / Widrow-Hoff learning rule)

Explanation: Unlike the original perceptron which updated weights based on quantized binary step outputs, ADALINE updates weights based on continuous linear errors.

Q63. How does the Backpropagation algorithm compute parameter gradients in a Multi-Layer Feedforward Network? [Basics of Neural Networks | Hard]

A) By applying the differential Calculus Chain Rule backward from the output layer through hidden layers

B) By brute-force random guessing of every weight matrix

C) By flipping binary weights alternately in forward passes

D) By applying Fourier transforms to input data

✓ Correct Answer: By applying the differential Calculus Chain Rule backward from the output layer through hidden layers

Explanation: Backpropagation calculates partial derivatives of the loss function with respect to each weight using the chain rule, enabling gradient descent updates.

Q64. In a Convolutional Neural Network (CNN), what is the primary role of Pooling (e.g. Max Pooling) layers? [Basics of Neural Networks | Medium]

A) To downsample spatial feature map dimensions, reduce computational parameters, and provide translation invariance

B) To increase image resolution and pixel depth

C) To convert 2D images directly into text labels without matrix operations

D) To generate negative weights for regularization

✓ Correct Answer: To downsample spatial feature map dimensions, reduce computational parameters, and provide translation invariance

Explanation: Max pooling extracts the maximum value within localized windows, reducing spatial dimensions while retaining salient feature responses.

Q65. What foundational architectural mechanism allows Transformers to process sequential data in parallel and capture long-range contextual relationships? [Basics of Neural Networks | Hard]

A) Self-Attention Mechanism (Query, Key, Value dot-product attention)

B) Sequential recurrent hidden loops like standard RNNs

C) Greedy search pruning of tree branches

D) Random dropout of 95% of neurons in every layer

✓ **Correct Answer: Self-Attention Mechanism (Query, Key, Value dot-product attention)**

Explanation: Transformers replace sequential recurrence with multi-head self-attention: Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) V.

UNIT VI: ADVANCED LEARNING TOPICS

Q66. What is Representation Learning in modern Machine Learning? [Representation Learning | Medium]

A) Learning feature representations directly from raw data that make it easier to extract useful information when building classifiers or predictors

B) Drawing manual flowchart representations of code logic

C) Hardcoding hand-crafted mathematical rules for each dataset feature

D) Creating UML class diagrams for software architecture

✓ **Correct Answer: Learning feature representations directly from raw data that make it easier to extract useful information when building classifiers or predictors**

Explanation: Deep neural networks automate feature engineering by learning hierarchical representations (low-level edges to high-level abstract concepts).

Q67. What is an Autoencoder in unsupervised representation learning? [Representation Learning | Medium]

A) A neural network trained to reconstruct its input through a bottleneck layer (encoder + latent space + decoder)

B) An automated bot that generates unit tests for code

C) A linear regression model that predicts continuous housing prices

D) A hardware compiler for GPUs

✓ **Correct Answer: A neural network trained to reconstruct its input through a bottleneck layer (encoder + latent space + decoder)**

Explanation: Autoencoders force the network to discover compressed latent representations of data by minimizing input-reconstruction error.

Q68. What is the core premise of Active Learning? [Active Learning | Medium]

A) The learning algorithm actively queries an oracle (e.g. human expert) to label specific unlabeled instances where uncertainty is highest

B) The computer screen remains active and never sleeps during training

C) The model only trains while the user actively types on the keyboard

D) Running machine learning algorithms on active server clusters

✓ **Correct Answer: The learning algorithm actively queries an oracle (e.g. human expert) to label specific unlabeled instances where uncertainty is highest**

Explanation: Active learning maximizes model accuracy with minimal labeling costs by selectively querying labels for the most informative or ambiguous data points.

Q69. What defines an Instance-Based (or Memory-Based / Lazy) learner such as k-Nearest Neighbors (k-NN)? [Instance-Based Learning | Medium]

A) It defers generalization and model building until a query or test instance needs to be evaluated

B) It precomputes a compact global parametric mathematical equation during training

C) It generates synthetic labeled instances to double training size

D) It requires retraining weights through thousands of backpropagation epochs

✓ **Correct Answer: It defers generalization and model building until a query or test instance needs to be evaluated**

Explanation: Instance-based learners simply store the training instances; computation occurs at prediction time by evaluating distances to stored exemplars.

Q70. What is Bagging (Bootstrap Aggregating) in ensemble learning, and which popular algorithm is based on it?

[Ensemble Learning | Medium]

A) Training multiple base models on bootstrap subsets of data with replacement and averaging predictions; Random Forest is a prominent example

B) Training models sequentially to correct previous errors; AdaBoost is an example

C) Combining models by training a meta-model on their output probabilities; Stacking is an example

D) Training a single deep neural network with 100 hidden layers

✓ Correct Answer: Training multiple base models on bootstrap subsets of data with replacement and averaging predictions; Random Forest is a prominent example

Explanation: Bagging reduces variance by aggregating predictions of diverse models trained on independent bootstrap samples of the training data.

Q71. What is Boosting in ensemble machine learning, and how does it differ from Bagging? [Ensemble Learning | Hard]

A) Boosting trains models sequentially where each subsequent learner focuses on correcting errors made by previous models, primarily reducing bias

B) Boosting trains models in parallel completely independent of one another

C) Boosting randomly prunes 50% of the training features before classification

D) Boosting requires unsupervised clustering before running linear regression

✓ Correct Answer: Boosting trains models sequentially where each subsequent learner focuses on correcting errors made by previous models, primarily reducing bias

Explanation: Algorithms like AdaBoost and Gradient Boosting sequentially adjust sample weights or fit residual errors to convert weak learners into a strong learner.

Q72. What is the difference between L1 Regularization (Lasso) and L2 Regularization (Ridge)? [Regularization Algorithms | Hard]

A) L1 adds the sum of absolute weights (|w|) driving parameters to exact zero (sparsity); L2 adds sum of squared weights (w^2) shrinking weights smoothly

B) L1 is used only for neural networks; L2 is used only for decision trees

C) L1 increases model overfitting; L2 eliminates bias completely

D) L1 always requires non-differentiable step activations

✓ Correct Answer: L1 adds the sum of absolute weights (|w|) driving parameters to exact zero (sparsity); L2 adds sum of squared weights (w^2) shrinking weights smoothly

Explanation: Lasso (L1) yields sparse models acting as automated feature selection; Ridge (L2) penalizes large weights preventing multicollinearity issues.

Q73. What is Dropout in deep neural network regularization? [Regularization Algorithms | Medium]

A) Randomly deactivating (setting to zero) a fraction p of hidden units during each forward training step to prevent co-adaptation of features

B) Dropping invalid rows from a CSV database during data cleaning

C) Stopping model training when CPU temperature exceeds 80 degrees

D) Deleting corrupted weights from hard disk storage

✓ Correct Answer: Randomly deactivating (setting to zero) a fraction p of hidden units during each forward training step to prevent co-adaptation of features

Explanation: Dropout forces the neural network to learn redundant and robust representations because no single neuron can rely solely on adjacent neurons.

Q74. What are the two competing neural networks that compose a Generative Adversarial Network (GAN)? [Generative Models | Medium]

A) The Generator (creates realistic synthetic data from random noise) and the Discriminator (distinguishes real data from synthetic fake data)

B) The Encoder and Decoder

C) The Actor and Critic

D) The Feedforward network and Recurrent network

✓ Correct Answer: The Generator (creates realistic synthetic data from random noise) and the Discriminator (distinguishes real data from synthetic fake data)

Explanation: GANs are framed as a two-player minimax game where the Generator tries to fool the Discriminator, while the Discriminator tries to detect fakes.

Q75. What is a Diffusion Model in generative machine learning? [Generative Models | Hard]

A) A generative model that gradually adds Gaussian noise to data in a forward process, and learns a neural network to reverse the denoising process to synthesize data

B) A model that spreads database queries across multiple network nodes

C) A clustering technique based on chemical diffusion rates

D) An algorithm that calculates shortest paths through optical fiber

✓ Correct Answer: A generative model that gradually adds Gaussian noise to data in a forward process, and learns a neural network to reverse the denoising process to synthesize data

Explanation: Modern state-of-the-art image generators (e.g., Stable Diffusion, Imagen) use reverse diffusion processes to generate high-fidelity samples from pure noise.

Q76. What role does the 'Discount Factor' (gamma, where 0 <= gamma <= 1) play in Reinforcement Learning? [Reinforcement Learning | Medium]

A) It determines the present value of future rewards, balancing immediate vs delayed rewards

B) It determines the learning rate step size during gradient descent

C) It measures the classification error on validation data

D) It specifies the maximum number of layers in a neural network

✓ Correct Answer: It determines the present value of future rewards, balancing immediate vs delayed rewards

Explanation: A gamma near 0 makes the agent myopic (greedy for immediate rewards), while a gamma near 1 makes the agent farsighted, valuing long-term cumulative rewards.

Q77. In Deep Q-Networks (DQN), what is the purpose of the 'Experience Replay' buffer? [Reinforcement Learning | Hard]

A) Storing past transitions (s, a, r, s') and sampling random mini-batches to break temporal correlation between consecutive experiences and stabilize training

B) Playing video game recordings to human users for entertainment

C) Storing trained weight parameters on secondary backup hard drives

D) Buffering network socket communication between client and server

✓ Correct Answer: Storing past transitions (s, a, r, s') and sampling random mini-batches to break temporal correlation between consecutive experiences and stabilize training

Explanation: Consecutive transitions in RL are highly correlated, leading to unstable gradient updates; uniform random sampling from replay memory breaks these correlations.

UNIT V: UNSUPERVISED LEARNING & NEURAL NETWORKS

Q78. What is the computational time complexity of the standard Lloyd's k-Means algorithm per iteration with n data points, d dimensions, and k clusters? [Clustering & Partition Methods | Hard]

A) $O(n \cdot k \cdot d)$

B) $O(n^2 \cdot d)$

C) $O(k^3 \cdot n)$

D) $O(d^2 \cdot \log n)$

✓ Correct Answer: $O(n \cdot k \cdot d)$

Explanation: In each iteration of k-Means, every one of the n points calculates Euclidean distance to k centroids across d dimensions.

Q79. What is the k-Means++ initialization algorithm designed to solve? [Clustering & Partition Methods | Medium]

A) Poor clustering convergence caused by arbitrarily choosing initial centroids too close to each other

B) Overfitting on large training datasets

C) Vanishing gradients in neural network backpropagation

D) Inability of k-Means to handle continuous numerical attributes

✓ **Correct Answer: Poor clustering convergence caused by arbitrarily choosing initial centroids too close to each other**

Explanation: k-Means++ selects the first centroid uniformly at random, then chooses subsequent centroids with probability proportional to the squared distance from the closest existing centroid.

Q80. In hierarchical agglomerative clustering, what is 'Single Linkage' (Minimum Linkage) and what artifact is it susceptible to? [Hierarchical Clustering | Hard]

A) Distance between two clusters is the minimum distance between any point in Cluster 1 and any point in Cluster 2; susceptible to 'Chaining' (elongated clusters)

B) Distance between cluster means; susceptible to spherical bias

C) Distance between cluster variances; susceptible to outliers

D) Distance computed using random sampling; susceptible to non-convergence

✓ **Correct Answer: Distance between two clusters is the minimum distance between any point in Cluster 1 and any point in Cluster 2; susceptible to 'Chaining' (elongated clusters)**

Explanation: Single linkage merges clusters linked by thin chains of intermediate points, often generating extended string-like clusters rather than compact groups.

Q81. What objective does Ward's Hierarchical Clustering method optimize at each merge step? [Hierarchical Clustering | Hard]

A) Minimizing the total within-cluster variance increase (minimum sum of squared differences from the centroid)

B) Maximizing the geometric distance between farthest points

C) Maximizing the total number of clusters

D) Minimizing the number of dimensions in the dataset

✓ **Correct Answer: Minimizing the total within-cluster variance increase (minimum sum of squared differences from the centroid)**

Explanation: Ward's minimum variance criterion merges the pair of clusters that minimizes the increase in total within-cluster sum of squares.

Q82. What does OPTICS (Ordering Points To Identify the Clustering Structure) improve over standard DBSCAN? [Density-Based Clustering | Hard]

A) It handles datasets with varying densities by producing an augmented cluster-ordering reachability plot rather than relying on a single fixed global epsilon

B) It runs in $O(1)$ constant time

C) It replaces distance metrics with decision trees

D) It requires user labels for cluster seeds

✓ **Correct Answer: It handles datasets with varying densities by producing an augmented cluster-ordering reachability plot rather than relying on a single fixed global epsilon**

Explanation: Standard DBSCAN struggles when clusters have significantly different local densities; OPTICS generates reachability plots to identify multi-density clusters.

Q83. What is the FP-Growth (Frequent Pattern Growth) algorithm and why is it faster than Apriori? [Pattern Finding & Association Rules | Hard]

A) It constructs a compact FP-Tree data structure in memory and uses recursive divide-and-conquer without generating candidate itemsets

B) It uses quantum computing gates to search databases

C) It only scans the database for single items

D) It discards 90% of transaction data at random

✓ **Correct Answer: It constructs a compact FP-Tree data structure in memory and uses recursive divide-and-conquer without generating candidate itemsets**

Explanation: FP-Growth avoids the exponential bottleneck of candidate generation by compressing transactions into a prefix tree and mining conditional pattern bases.

Q84. What is the mathematical threshold logic function computed by the original 1943 McCulloch-Pitts artificial neuron? [Basics of Neural Networks | Medium]

A) $y = 1$ if $\sum w_i x_i \ge \theta$ and no inhibitory inputs are active; otherwise $y = 0$

B) $y = \frac{1}{1 + e^{-z}}$

C) $y = \max(0, z)$

D) $y = \tanh(z)$

✓ Correct Answer: $y = 1$ if $\sum w_i x_i \ge \theta$ and no inhibitory inputs are active; otherwise $y = 0$

Explanation: The McCulloch-Pitts neuron modelled biological excitation and absolute inhibition using binary threshold logic with fixed integer weights.

Q85. What is the Perceptron Convergence Theorem formulated by Frank Rosenblatt? [Basics of Neural Networks | Hard]

A) If the training dataset is linearly separable, the Perceptron learning algorithm is guaranteed to converge to a separating hyperplane in a finite number of steps

B) Any neural network with 3 layers can learn any continuous function

C) Perceptrons converge faster when learning rates are infinite

D) Perceptrons can separate non-linear circles without kernel transformations

✓ Correct Answer: If the training dataset is linearly separable, the Perceptron learning algorithm is guaranteed to converge to a separating hyperplane in a finite number of steps

Explanation: Rosenblatt proved that given linear separability and bounded inputs, error-correction weight updates will terminate with zero training error.

Q86. What is the derivative of the standard Sigmoid function $\sigma(z)$ expressed in terms of its output? [Basics of Neural Networks | Hard]

A) $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

B) $\sigma'(z) = 1 - \sigma(z)^2$

C) $\sigma'(z) = \max(0, 1)$

D) $\sigma'(z) = \frac{1}{\sigma(z)}$

✓ Correct Answer: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Explanation: Because its derivative has maximum value 0.25 (at $z=0$), chaining gradients across multiple layers quickly leads to vanishing gradients ($0.25^L \to 0$).

Q87. What is the Dying ReLU problem in deep neural network training? [Basics of Neural Networks | Medium]

A) If a large gradient updates weights such that the neuron produces negative values for all training samples, its gradient is permanently 0 and the neuron never activates again

B) The GPU runs out of video memory during backpropagation

C) The loss function oscillates to infinity

D) The network loses connection to the training server

✓ Correct Answer: If a large gradient updates weights such that the neuron produces negative values for all training samples, its gradient is permanently 0 and the neuron never activates again

Explanation: Because the derivative of ReLU is 0 for $z < 0$, weights cannot recover once a neuron gets knocked into negative activation across the entire dataset.

Q88. How does Leaky ReLU resolve the Dying ReLU problem? [Basics of Neural Networks | Medium]

A) By introducing a small, non-zero slope (e.g. $\alpha = 0.01$) for negative inputs ($f(x) = \alpha x$ when $x < 0$)

B) By inverting positive values to negative values

C) By converting all weights to complex numbers

D) By resetting weights to zero after every epoch

✓ Correct Answer: By introducing a small, non-zero slope (e.g. $\alpha = 0.01$) for negative inputs ($f(x) = \alpha x$ when $x < 0$)

Explanation: Leaky ReLU ensures a small gradient always flows backward even when activations are negative, preventing dead units.

Q89. What is a Competitive Neural Network (such as a Kohonen Self-Organizing Map / SOM)? [Basics of Neural Networks | Hard]

A) An unsupervised network where neurons compete for the right to respond to input patterns using a Winner-Takes-All rule

B) A network where two players play chess against each other

C) A network that optimizes linear programming models

D) A supervised classifier using hinge loss

✓ Correct Answer: An unsupervised network where neurons compete for the right to respond to input patterns using a Winner-Takes-All rule

Explanation: In competitive learning, the neuron whose weight vector is closest to the input vector wins and adapts its weights and neighborhood towards the input.

Q90. What is the Exploding Gradient problem in Recurrent Neural Networks (RNNs) and how is it standardly mitigated? [Basics of Neural Networks | Medium]

A) Gradients accumulate exponentially through unrolled time steps causing numeric overflow; mitigated using Gradient Clipping (capping maximum gradient norm)

B) Gradients diminish to zero; mitigated by adding more layers

C) Loss becomes negative; mitigated by removing hidden layers

D) Weights flip signs alternately; mitigated by multiplying by -1

✓ Correct Answer: Gradients accumulate exponentially through unrolled time steps causing numeric overflow; mitigated using Gradient Clipping (capping maximum gradient norm)

Explanation: Gradient clipping rescales gradient vectors when their L_2 norm exceeds a threshold c: $g \leftarrow c \frac{g}{\|g\|}$.

Q91. What are the three gates present in a Long Short-Term Memory (LSTM) cell? [Basics of Neural Networks | Medium]

A) Forget Gate, Input Gate, and Output Gate

B) Read Gate, Write Gate, and Execute Gate

C) Encode Gate, Latent Gate, and Decode Gate

D) AND Gate, OR Gate, and XOR Gate

✓ Correct Answer: Forget Gate, Input Gate, and Output Gate

Explanation: LSTMs regulate internal cell state C_t via sigmoid gates: forget gate (what to discard), input gate (what new information to store), output gate (what to emit).

Q92. What is the difference between standard convolution and 1x1 Convolution (Pointwise Convolution) in deep CNN architectures (like Inception)? [Basics of Neural Networks | Hard]

A) 1x1 convolution acts as cross-channel pooling, reducing or increasing feature map channel depth without altering spatial width and height

B) 1x1 convolution downsamples spatial resolution by half

C) 1x1 convolution completely removes spatial channels

D) 1x1 convolution can only be applied to black-and-white photos

✓ Correct Answer: 1x1 convolution acts as cross-channel pooling, reducing or increasing feature map channel depth without altering spatial width and height

Explanation: 1x1 convolutions compute linear combinations across input channels followed by non-linearities, allowing dimensionality reduction before expensive 3x3/5x5 convolutions.

Q93. What is Positional Encoding in Transformer models? [Basics of Neural Networks | Hard]

A) Vector representations added to token embeddings to inject sequence order information into the permutation-invariant self-attention layers

B) GPS tracking coordinates embedded in mobile devices

C) Index numbers assigned to hidden layer weights

D) Labels assigned to tokens by human annotators

✓ Correct Answer: Vector representations added to token embeddings to inject sequence order information into the permutation-invariant self-attention layers

Explanation: Because multi-head attention operates on sets of tokens simultaneously without recurrence, sinusoidal or learned positional embeddings restore word order.

UNIT VI: ADVANCED LEARNING TOPICS

Q94. What is Self-Supervised Learning (SSL) and how does it relate to Representation Learning? [Representation Learning | Hard]

A) The model creates its own supervisory signals from unlabeled data (e.g. predicting masked words or contrasting augmented image views) to learn rich feature representations

B) A human supervisor writes rules for the model once a day

C) The model trains exclusively without loss functions

D) A technique where models evaluate their own accuracy on test sets

✓ Correct Answer: The model creates its own supervisory signals from unlabeled data (e.g. predicting masked words or contrasting augmented image views) to learn rich feature representations

Explanation: Pre-training paradigms like BERT (Masked Language Modeling) and SimCLR (Contrastive Learning) extract generalizable representations from massive unlabeled corpora.

Q95. Which query strategy in Active Learning selects unlabeled instances where the difference between the top two highest predicted class probabilities is smallest? [Active Learning | Hard]

A) Margin Sampling

B) Least Confidence Sampling

C) Entropy Sampling

D) Random Uniform Sampling

✓ Correct Answer: Margin Sampling

Explanation: Margin sampling targets data points where the classifier is most ambiguous between the top two competitor classes: $x^* = \arg\min_x (P(\hat{y}_1|x) - P(\hat{y}_2|x))$.

Q96. What is the 'Curse of Dimensionality' in high-dimensional instance-based learning (k-NN)? [Instance-Based Learning | Medium]

A) As the number of features increases, data points become equidistant from one another in high-dimensional space, rendering distance metrics uninformative

B) High dimensional matrices cannot be multiplied on modern GPUs

C) The number of classes grows exponentially with dataset size

D) k-NN cannot be run on datasets with more than 5 columns

✓ Correct Answer: As the number of features increases, data points become equidistant from one another in high-dimensional space, rendering distance metrics uninformative

Explanation: In high dimensions, the ratio between the distance to the nearest neighbor and the distance to the farthest neighbor approaches 1, diminishing neighbor distinctions.

Q97. What is the mathematical definition of Stacking (Stacked Generalization) in ensemble learning? [Ensemble Learning | Medium]

A) Training multiple heterogeneous base models (e.g. SVM, Random Forest, Neural Net) and training a Meta-Learner on their combined prediction outputs

B) Summing all decision tree weights without training

C) Stacking datasets on top of each other in a spreadsheet

D) Running the same model 10 times in a loop

✓ Correct Answer: Training multiple heterogeneous base models (e.g. SVM, Random Forest, Neural Net) and training a Meta-Learner on their combined prediction outputs

Explanation: Stacking combines predictions of diverse model families via out-of-fold cross-validation to prevent target leakage into the meta-model.

Q98. In AdaBoost (Adaptive Boosting), how are the weights of training instances modified after each weak hypothesis is trained? [Ensemble Learning | Medium]

A) Weights of incorrectly classified instances are increased, forcing subsequent weak learners to concentrate on hard examples

B) Weights of all instances are reset to zero

C) Weights of correctly classified instances are doubled

D) Incorrect instances are deleted from the training set

✓ Correct Answer: Weights of incorrectly classified instances are increased, forcing subsequent weak learners to concentrate on hard examples

Explanation: AdaBoost updates sample weights with $w_i \leftarrow w_i \exp(\alpha_m \cdot \mathbb{I}(y_i \ne h_m(x_i)))$, focusing attention on misclassified points.

Q99. How does Gradient Boosting differ fundamentally from AdaBoost? [Ensemble Learning | Hard]

A) Gradient Boosting optimizes any differentiable loss function by training base learners to fit pseudo-residuals (negative gradients) rather than re-weighting data points

B) Gradient Boosting cannot be used for regression

C) Gradient Boosting trains all trees in parallel

D) Gradient Boosting does not use decision trees

✓ Correct Answer: Gradient Boosting optimizes any differentiable loss function by training base learners to fit pseudo-residuals (negative gradients) rather than re-weighting data points

Explanation: Gradient boosting views boosting as gradient descent in function space, sequentially fitting new base models to the residual errors of the current ensemble.

Q100. What is Early Stopping in iterative machine learning algorithms? [Regularization Algorithms | Easy]

A) Halting model training when performance on an independent validation set begins to deteriorate, preventing overfitting

B) Terminating the server when training exceeds 1 hour

C) Stopping the test set evaluation after 10 samples

D) Deleting model checkpoints periodically

✓ Correct Answer: Halting model training when performance on an independent validation set begins to deteriorate, preventing overfitting

Explanation: Early stopping acts as a regularizer by limiting effective parameter capacity and stopping gradient updates before the model fits noise.

Q101. What is Batch Normalization and what are its primary benefits during deep network training? [Regularization Algorithms | Hard]

A) Normalizing layer inputs across each mini-batch (zero mean, unit variance) followed by learnable scaling and shifting (gamma and beta); speeds up training and stabilizes gradients

B) Sorting training records in alphabetical order

C) Compressing mini-batches with gzip

D) Rounding all weight parameters to integer values

✓ Correct Answer: Normalizing layer inputs across each mini-batch (zero mean, unit variance) followed by learnable scaling and shifting (gamma and beta); speeds up training and stabilizes gradients

Explanation: Batch Normalization reduces internal covariate shift, allows significantly higher learning rates, and acts as a mild regularizer.

Q102. What is the Mode Collapse failure mode in Generative Adversarial Networks (GANs)? [Generative Models | Hard]

A) The Generator discovers a narrow subset of outputs that fool the Discriminator and produces limited, repetitive samples rather than diverse data distributions

B) The Generator runs out of memory and crashes

C) The Discriminator achieves 100% loss on real images

D) The generator and discriminator swap weight parameters

✓ Correct Answer: The Generator discovers a narrow subset of outputs that fool the Discriminator and produces limited, repetitive samples rather than diverse data distributions

Explanation: Mode collapse occurs when the generator maps multiple latent vectors to the same small set of realistic outputs, failing to capture full data variety.

Q103. What loss component in Variational Autoencoders (VAEs) measures the difference between the approximate posterior distribution $q_\phi(z|x)$ and the prior $p(z)$? [Generative Models | Hard]

A) Kullback-Leibler (KL) Divergence

B) Mean Squared Error (MSE)

C) Binary Cross Entropy

D) Hinge Loss

✓ Correct Answer: Kullback-Leibler (KL) Divergence

Explanation: The VAE loss combines reconstruction loss (measuring data fidelity) and KL divergence (regularizing latent representations towards standard normal distributions).

Q104. What is the Bellman Optimality Equation for the state-value function $V^*(s)$ in dynamic programming / RL?
[Reinforcement Learning | Hard]

A) $V^*(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma V^*(s')]$

B) $V^*(s) = \sum_a \pi(a|s) Q(s, a)$

C) $V^*(s) = r + \gamma$

D) $V^*(s) = \frac{1}{1 - \gamma}$

✓ Correct Answer: $V^*(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma V^*(s')]$
Explanation: The Bellman optimality equation states that the value of an optimal state equals the expected immediate reward plus discounted optimal value of the successor state.

Q105. What is the difference between On-Policy algorithms (like SARSA) and Off-Policy algorithms (like Q-Learning)?
[Reinforcement Learning | Hard]

A) On-Policy evaluates and improves the exact policy currently used to take actions; Off-Policy learns an optimal target policy independently of the exploratory behavior policy

B) On-Policy runs on real robots; Off-Policy runs in software simulations

C) On-Policy requires neural networks; Off-Policy uses lookup tables only

D) On-Policy does not use discount factors

✓ Correct Answer: On-Policy evaluates and improves the exact policy currently used to take actions; Off-Policy learns an optimal target policy independently of the exploratory behavior policy
Explanation: SARSA updates using action a' chosen by its current epsilon-greedy policy: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)]$, while Q-learning uses $\max_{a'} Q(s', a')$.

Q106. In Policy Gradient methods (such as REINFORCE), what is directly parameterized by the neural network weights θ? **[Reinforcement Learning | Hard]**

A) The policy distribution $\pi_\theta(a|s)$ mapping states directly to action selection probabilities

B) The transition probability table of the environment

C) The discounted cumulative reward discount factor

D) The physical mass of the agent robot

✓ Correct Answer: The policy distribution $\pi_\theta(a|s)$ mapping states directly to action selection probabilities
Explanation: Policy gradient methods bypass value function discretization, allowing seamless operation in continuous action spaces.
